

Test-Time Physical Awareness for Generated Humanoid Reference Motions

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Abstract

Modern humanoid motion trackers such as SONIC can follow rich whole-body reference trajectories in physics, but those trackers still need high-quality reference motions. KIMODO makes reference authoring much easier by generating Unitree G1 kinematic motions from text prompts and constraints. This project studies the interface between those systems: given a generated G1 reference, can we detect physical risks, use those risks to rewrite the prompt, regenerate better candidates, and then gate the result with a controller rollout?

I built a test-time physical-awareness loop. It evaluates generated references with MuJoCo inverse-dynamics/contact diagnostics and a pretrained SONIC controller audit. The evaluator emits interpretable failure tags such as high torque/root wrench, self-contact, non-foot floor contact, weak support, contact artifact, and controller trackability failure. Those tags are passed as structured context to GPT-5.5, which rewrites the prompt with a metric-specific safety constraint. On the 100-prompt KIMODO benchmark, first-pass references pass the physical screen in 48/100 cases, complete the 4-second SONIC rollout in 53/100 cases, and pass both gates in 29/100 cases. The final deck reports a prompt-repair snapshot where metric-guided rewriting reduces several failure tags, including SONIC fall cases from 47 to 20. The contribution is not a solved text-to-robot motion generator; it is a concrete evaluation, prompt-repair, and acceptance gate for exposing generated-reference brittleness.

1 Problem

Humanoid motion control has recently improved because learned policies can be trained to track diverse reference motions under physics. SONIC is a strong example: it trains a Unitree G1 whole-body controller on over 100M frames / 700 hours of high-quality human motion [1]. At the input level, the tracker receives a target trajectory: root pose and joint targets over time.

The upstream question is where those references come from. KIMODO generates controllable kinematic human and robot motions from text prompts and constraints [2]. This makes it a natural source of candidate references for robot training and controller testing. However, a kinematic reference is not automatically physically executable. It may demand unrealistic torques, require impossible support, create self-collisions, drag contacts, or destabilize a learned controller.

The project question is:

Can a physical-awareness layer identify risky KIMODO-generated humanoid reference motions, and can its failure tags guide useful test-time prompt repair before controller execution?

2 System

The final system is a prompt-refinement loop:

Stage	Input	Contribution	Output
Generate	text prompt	KIMODO G1 generation	root + joint trajectory
Evaluate	reference	MuJoCo + SONIC checks	risk flags and rollout metrics
Rewrite	failed metric tag	GPT-5.5 prompt edit	revised prompt
Repeat/gate	new candidate	rescore until budget ends	accepted or flagged clip

Failure tags are intentionally interpretable. High torque/root wrench asks the LLM for slower, smoother limb motion; self-contact asks for limbs away from the torso; weak support asks for wider stance; trackability failure asks for lower speed and amplitude. SONIC is used as a foundation-style tracking evaluator and acceptance gate. It is not the optimizer that edits KIMODO.

3 Benchmark and Metrics

The benchmark contains 100 humanoid robotics prompts grouped as locomotion and recovery (26), manipulation and loco-manipulation (26), floor/low posture (12), dance/expressive motion (12), athletic and terrain stress (16), and communication/safety gestures (8). The suite includes walking, shuffling, crawling, carrying, panel pressing, gestures, obstacle-style steps, forward rolls, cartwheel attempts, jumping, and low-posture motions.

For each generated G1 trajectory, the evaluator computes torque demand relative to actuator limits, root force/torque, velocity, acceleration, jerk, self-contact, non-foot floor contact, contact artifacts, support-region proxy violations, SONIC no-fall, survival time, and joint RMSE.

4 Results

4.1 First-Pass Audit

Table 1: First-pass 100-prompt KIMODO audit.

Set	Physical pass	SONIC no-fall	Both	Mean sec.	RMSE
All KIMODO refs	48/100	53/100	29/100	2.855	0.156
Physical-pass subset	48/48	29/48	29/48	3.293	0.170
Flagged subset	0/52	24/52	0/52	2.450	0.143

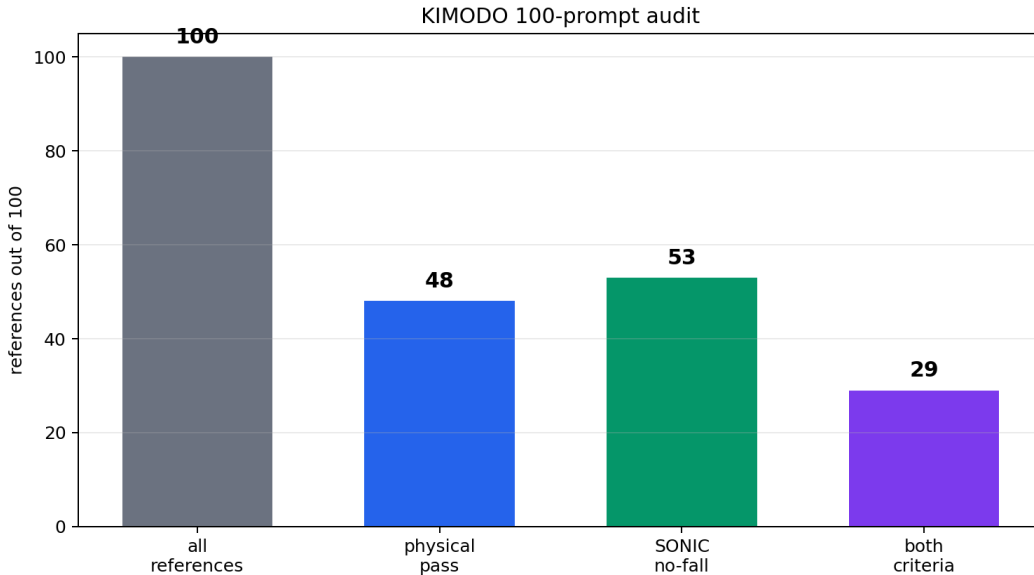


Figure 1: The physical screen and SONIC audit are related but not equivalent.

Only 29 of 48 physical-pass references also complete SONIC. Meanwhile, 24 of 52 flagged references still complete SONIC. This disagreement is a central result: reference-level physical checks and controller-level trackability are both needed.

4.2 Failure Tags

Table 2: Failure-tag counts before and after metric-guided prompt repair. Tags are non-exclusive.

Failure tag	Before	After
SONIC fall	47	20
Torque limit >1x	48	23
Self-contact >8%	34	10
Unstable center of mass	31	20
High root force >5 kN	28	9
Non-foot floor contact	11	4

The representative failure videos in the deck are ‘forward_roll’ for torque/root wrench, ‘turn_in_place’ for self-contact, ‘knee_slide’ for non-foot floor contact, ‘step_over_right’ for support proxy, ‘carry_box’ for controller trackability, and ‘happy_dance’ for contact artifact.

4.3 Repair Examples

The main repair videos show failure-tag-guided prompt edits: ‘forward_walk’ becomes slower and smoother after a trackability flag; ‘wave’ adds planted feet and smooth arm motion after torque/tracking flags; ‘split_squat_jump’ asks for a smaller jump and soft landing after support/torque flags; ‘pushup_pose’ asks for lifted torso clearance; ‘happy_dance’ asks for clean foot placement; and ‘backward_recovery’ asks for a smaller lean and step back.

5 Interpretation

The results support the hypothesis in a bounded way. Generated references have measurable physical failure modes; failure tags are useful for prompt refinement and curation; SONIC is stricter than kinematic replay; and visual/VLM review helps catch obvious behavior mismatch, but it is not a complete metric set.

The boundary slide shows remaining failures: ‘forward_roll’ for torque/root wrench stress, ‘carry_box’ for controller trackability, and ‘cartwheel_attempt’ for acrobatics. The current method does not solve arbitrary text-to-robot motion, robust acrobatics/low-posture tracking, or automatic repair for every invalid reference. Stronger results likely require fine-tuning or training a motion-level refiner, not only prompt rewriting.

6 Limitations and Future Work

GPT-5.5 prompt repair is useful but cannot replace a better motion generator. The physical screen is heuristic and simulator-dependent. SONIC rollout is a simulation diagnostic, not real hardware validation. Some motions need semantic context: floor contact can be correct for crawling and wrong for walking. Prompt fidelity is checked qualitatively; there is no full human-study or VLM benchmark.

Future work should train a risk-conditioned refiner that learns from failure tags and SONIC rollouts, use short controller rollouts or a learned surrogate during candidate selection, and improve semantic evaluation so physical stability does not hide prompt mismatch.

7 Team Responsibilities

Tae Hoon Yang completed the project solo: prompt suite design, KIMODO/SONIC integration, MuJoCo physical checks, prompt-repair loop, video rendering, slide deck, and final report.

References

- [1] Zhengyi Luo, Ye Yuan, Tingwu Wang, Chenran Li, Sirui Chen, Fernando Castañeda, Zi-Ang Cao, Jiefeng Li, David Minor, Qingwei Ben, Xingye Da, Runyu Ding, Cyrus Hogg, Lina Song, Edy Lim, Eugene Jeong, Tairan

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- [2] Davis Rempe, Mathis Petrovich, Ye Yuan, Haotian Zhang, Xue Bin Peng, Yifeng Jiang, Tingwu Wang, Umar Iqbal, David Minor, Michael de Ruyter, Jiefeng Li, Chen Tessler, Edy Lim, Eugene Jeong, Sam Wu, Ehsan Hassani, Michael Huang, Jin-Bey Yu, Chaeyeon Chung, Lina Song, Olivier Dionne, Jan Kautz, Simon Yuen, and Sanja Fidler. Kimodo: Scaling controllable human motion generation, 2026.